KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Answers · Term 3

TERM 3

Teacher copy. Answers were calculated in code and checked. Circumference and Pythagoras answers use the π and √ keys; where π ≈ 3.14 gives a different 1 d.p. answer it is listed as an alternative. Reasoning answers are model responses: accept any correct explanation.

WK 1	Q1	Q2	Q3	Q4	Q5
S1	$w = 3$	$3m^2 + 7m$	$x = 12$ cm	−2	$\frac{3}{8}$
S2	Yes: both sides equal 30 when $m = 3$.	40%	−4	256 m^3	4
S3	$m = 2$ or $m = -2$	$25a + 35$	60 000 m^2	$\frac{3}{5}$ (0.6)	9^2

WK 2	Q1	Q2	Q3	Q4	Q5
S1	$5n - 1 = 9$; the number is 2	4^8	25.1 m	$x = 123$	$7(x - 1)$
S2	$n = 7$	$30	$x = 54$	96 cm	$20m^2$
S3	$n = 8$	200 kg	Yes: when $x = 1$, $y = -3$.	$2m + 20$	48 m^3

WK 3	Q1	Q2	Q3	Q4	Q5
S1	$m = 1.2$ (= $1\frac{1}{5}$)	$8k - 12$	$x = 5$ cm	12	Mia multiplied 3 × 4; 3^4 means 3 × 3 × 3 × 3 = 81.
S2	40% profit	3^9	11	$k = 4$	27 cm^2
S3	$m = 6$ or $m = -6$	$5(2y - 5)$	$x = 4$ cm	30 L	$x = 108$

WK 4	Q1	Q2	Q3	Q4	Q5
S1	$n = 4$	105 mL	$2^3 \times 5^2$	55 m^2	$\frac{2}{7}$
S2	Mia only multiplied the first term; both terms must be multiplied by 6: $6p + 48$.	6	40 m^2	$5n - 12 = 3$; the number is 3	$x = 129$
S3	$3n + 8$	$k = 6$	Yes: $15^2 + 20^2 = 625 = 25^2$	$x = 39$	$24

WK 5	Q1	Q2	Q3	Q4	Q5
S1	3^4	$x = 5.4$ cm	$y = 7$ or $y = -7$	$0.\dot{4}$	$x = 63$
S2	$8(3y + 8)$	5^8	149 cm	30 m^2	$h = 6$ (hours)
S3	$4m^2 + 7m$	$x = 17$ cm	90%	$n = 2$	The rectangle: rectangle 50 cm^2, square 49 cm^2.

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WK 6	Q1	Q2	Q3	Q4	Q5
S1	$a = 0.6 \left(= \frac{3}{5}\right)$	5^9	$10a - 2$	No: P(red) = $\frac{6}{13}$, which is less than $\frac{1}{2}$.	576 m^3
S2	$x = 8$ m	$2^3 \times 7$	$x = 128$	7	$m = 2$ or $m = -2$
S3	−14	160	$6n$	$2n + 6 = 16$; the number is 5	69.1 cm

WK 7	Q1	Q2	Q3	Q4	Q5
S1	6 and 7 (because 36 < 45 < 49)	$3(3y + 1)$	0.00038	$n = 2$	5
S2	No: $5^2 + 6^2 = 61$, but $8^2 = 64$	$5x + 16$	1.2	$x = 112$ (angle sum of a quadrilateral is 360°)	$t = 3$
S3	$4p$	$m = 6$ or $m = -6$	$x = 145$	$13	45 m

WK 8	Q1	Q2	Q3	Q4	Q5
S1	$8k - 14$	25%	0.8	$x = 15$ cm	5
S2	$k = 4$	9^3	16 cm^2	$140	$\frac{1}{2}$ (0.5)
S3	Yes: both are $30 (50% of 60 = 60% of 50 = 50 × 60 ÷ 100).	$2(3y - 1)$	$x = 55$	750 mL	$m = 2.6 \left(= 2\frac{3}{5}\right)$

WK 9	Q1	Q2	Q3	Q4	Q5
S1	$w = 3$	$8a + 6$	$x = 30$ cm	No: substituting $k = 10$ does not make both sides equal (the solution is $k = 11$).	17.25
S2	$4k + 4$	3^{12}	72 cm^3	5	17 L
S3	$x = 6.4$ m	25% profit	$n = 9$	$x = 63$	$k = 9$

WK 10	Q1	Q2	Q3	Q4	Q5
S1	8^9	$y = 7$ or $y = -7$	$x = 25$ cm	$x = 53$	30
S2	$350	No: $4^2 + 6^2 = 52$, but $7^2 = 49$	$2n + 9 = 27$; the number is 9	$14m$	Sam added first; × must be done before +. Correct answer 105.
S3	$w = 2$	2	2	$x = 39$	$x = 12$ m